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Subject Code: MPH203T

Roll No:

MPHARM

(SEM. II) THEORY EXAMINATION 2025-26

COMPUTER AIDED DRUG DELIVERY SYSTEM

TIME: 3 HRS

M.MARKS: 75

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief.

10x2 = 20

- (a) Define computer-aided drug development.
- (b) What is statistical modeling?
- (c) Define Quality by Design (QbD).
- (d) What is sensitivity analysis?
- (e) Define factorial design.
- (f) What is P-gp transporter?
- (g) Define IVIVC.
- (h) What is virtual trial?
- (i) Define artificial intelligence.
- (j) What is pharmaceutical automation?

SECTION B

2. Attempt any *two* of the following:

2 x 10 = 20

- (a) Discuss history and applications of computers in pharmaceutical research and development.
- (b) Explain Quality by Design (QbD) in pharmaceutical development with reference to ICH Q8 guidelines.
- (c) Describe computational modeling techniques used in drug absorption, distribution and excretion.

SECTION C

3. Attempt any five part of the following:

7 x 5 = 35

- (a) Write a note on descriptive and mechanistic modeling.
- (b) Discuss optimization parameters used in formulation development.
- (c) Explain computer-aided development of pharmaceutical emulsions and microemulsions.
- (d) Write a short note on intestinal permeation models.
- (e) Discuss gastrointestinal absorption simulation.
- (f) Explain computer simulations in pharmacokinetics and pharmacodynamics.
- (g) Write a note on regulation of computerized systems in clinical development.